Installing and Running the Elastic Moduli GUI

Petrophysics Course | Student Guide

This guide explains how to create an isolated Conda environment, install the packages required by the graphical interface, and launch the Elastic Moduli GUI. Anaconda must already be installed before you begin.

Before you begin

Requirement	What you need
Software	Anaconda already installed
Files	GUI_V0.py and the experimental data file
Internet	Required only while installing the Python packages
Operating system	Windows or macOS

Create the Conda environment

1 Open a Conda enabled terminal

- Windows: open Anaconda Prompt from the Start menu.
- macOS: open Terminal. If conda is not recognized, open a terminal from Anaconda Navigator or initialize your shell as described in the troubleshooting section.

2 Create a new environment

Enter the following command (copy/paste):

```
conda create --name MechAcoustoAnalyzer python=3.11 numpy pandas matplotlib PyQt6
```

When Conda asks whether to proceed, type y and press Enter.

3 Activate the environment

```
conda activate MechAcoustoAnalyzer
```

The beginning of the terminal line should now show (MechAcoustoAnalyzer). This confirms that you are in the new environment rather than in the base environment.

Check the installation and launch the GUI

1 Check the installed packages

```
python -c "import numpy,pandas,matplotlib,PyQt6"
```

If the command finishes without an error message, the required packages are available.

2 Keep the program and data together

Create or download a folder with this structure:

```
Elastic_Moduli_GUI/
 GUI_V0.py
 b849_reduced.csv
```

If the reduced dataset is named b849_reduced.csv and is in the same folder as the program, the GUI loads it automatically. You can also open a different raw or reduced file from inside the GUI.

3 Move the terminal to the folder

Examples for a folder placed on the Desktop:

System	Command
Windows	`cd %USERPROFILE%\Desktop\Elastic_Moduli_GUI`
macOS	`cd ~/Desktop/Elastic_Moduli_GUI`

If your folder is elsewhere, type cd followed by a space, then enter or paste the actual folder path. Put quotation marks around paths that contain spaces.

4 Launch the program

```
python GUI_V1.py
```

To launch the GUI while explicitly specifying the dataset, use:

```
python GUI_V1.py b849_reduced.csv
```

First use

When the window opens:

- If no curve appears, select Open Data File (Raw / Reduced CSV) and choose the supplied dataset.
- For the supplied experiment, the calculations use a 38 mm diameter and 76 mm initial specimen length.
- Move q min and q max to select the differential stress interval used to fit Young's modulus.
- Inspect both E and R^2. A high R^2 supports linearity, but the selected interval must also be physically meaningful.
- Use Save Current E & R^2 Value to compare estimates from different intervals.

Using the GUI again

The environment and packages only need to be created once. For later sessions:

```
conda activate elastic-moduli

cd PATH_TO_THE_Elastic_Moduli_GUI_FOLDER

python GUI_V0.py
```

When you finish, close the GUI and deactivate the environment:

```
conda deactivate
```

Troubleshooting

1. conda is not recognized

Use Anaconda Prompt on Windows. On macOS, reopen Terminal after installing Anaconda. If needed, launch a terminal configured by Anaconda or run Conda shell initialization from the Anaconda installation.

2. No module named numpy pandas matplotlib or PyQt6

Confirm that (elastic-moduli) is visible at the start of the terminal line, then repeat the package installation command.

3. python cannot open GUI_V1.py

The terminal is not in the program folder. Use cd to move to the folder and use dir on Windows or ls on macOS to confirm that GUI_V1.py is present.

4. The GUI opens but no data appear

Place b849_reduced.csv beside GUI_V1.py, or select Open Data File (Raw / Reduced CSV) and choose the dataset manually.

5. A Qt or display error appears

With the environment active, reinstall the graphical package using python -m pip install --force-reinstall PyQt6, then launch the program again.

Installation checklist

- [ ] Anaconda is installed.
- [ ] The elastic-moduli environment exists and is active.
- [ ] NumPy, pandas, Matplotlib and PyQt6 are installed.
- [ ] GUI_V1.py and the dataset are in the expected folder.
- [ ] The GUI opens and displays the stress strain curve.

Learning focus The purpose of the tool is not only to obtain a numerical modulus. Compare intervals, evaluate linearity, and explain why the measured stiffness changes as the rock progresses from crack closure toward damage and failure.